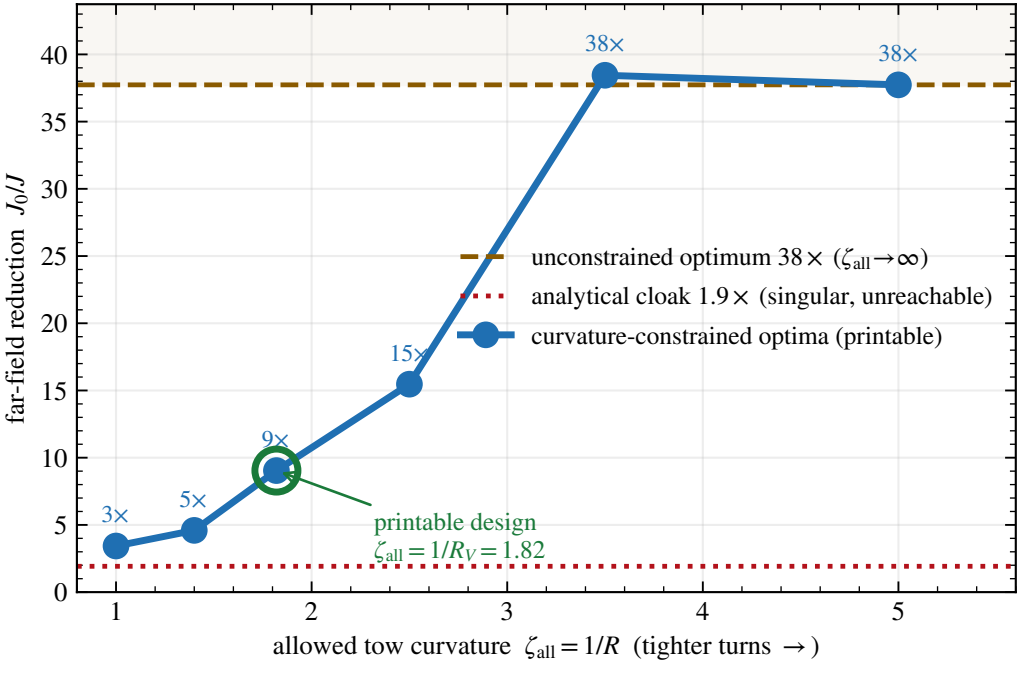
far-field reduction $J_0/J$
allowed tow curvature $\zeta_{\mathrm{all}} = 1/R$ (tighter turns →)
unconstrained optimum 38× ($\zeta_{\mathrm{all}} \to \infty$)
analytical cloak 1.9× (singular, unreachable)
curvature-constrained optima (printable)
3×
5×
9×
15×
38×
38×
printable design
$\zeta_{\mathrm{all}} = 1/R_V = 1.82$
0
5
10
15
20
25
30
35
40
1
2
3
4
5